

Alexandre Duteau

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EDUCATION

University of Calgary – Bachelor of Science in Computer Science

Expected Apr 2027

WORK EXPERIENCE

Software Engineer, Midas Labs – Calgary, AB

Jun 2024 – Present

- Engineered an embeddable AI chat widget deployed across multiple clients, owning the flagship build end to end for two years and serving 64,000+ queries for a credit counselling nonprofit
- Designed and shipped a serverless AWS backend in Terraform (API Gateway, Lambda, IAM, CloudWatch), stripping all third party credentials from shipped JavaScript; live at 250 req/day with 100% success
- Top contributor of five engineers on a Go/Postgres career coaching marketplace (56 REST endpoints, 37 migrations, 503 tests), including presigned S3 uploads, SES email delivery, and double booking prevention
- Shipped four Shadow DOM web components across hundreds of member associations on an association management SaaS, including a union based audience model for campaign targeting
- Built an AI email drafting experience on AWS Bedrock with streamed rendering, then fixed the merge token validation blocking every generated draft from reaching approved status

OTHER TECHNICAL EXPERIENCE

Co-Founder & Software Engineer, Nullus Inc. – Calgary, AB

Nov 2023 – Present

- Developed 8 client applications in Next.js, TypeScript, React, and Go on Vercel, plus an in progress React Native rewards app for a restaurant chain
- Drove adoption of a ticket-based PR workflow using Jira with issue-keyed branches, conventional commits, and mandatory peer review, saving the team 4+ hours weekly in rework and merge conflicts

Web Developer, Canadian Global Care on Campus – Calgary, AB

Oct 2023 – Apr 2025

- Owned the club's public site across multiple event cycles, serving 200+ weekly visitors at peak
- Rebuilt it from a static page into a 5 route React and Vite app, shipping 2 weeks early and lifting event attendance 36%

PROJECTS

NFL Game Prediction Pipeline – Python, XGBoost, Scikit-learn, Pandas, Flask, Next.js

- Built a scraping system across 28 scripts using BeautifulSoup, requests, and Selenium to harvest 1,673 NFL games (2018 to 2024) into 30,000+ structured CSVs, handling paginated navigation, inconsistent table schemas, and rate limiting
- Trained an XGBoost classifier to 71% accuracy against a 62.8% majority class baseline using class weight balancing, SMOTE oversampling, and decision threshold optimization, validated with a walk-forward backtest across the 2024 season that retrained weekly on prior data only to avoid look ahead bias

UrbanSignal – Next.js, React, TypeScript, Supabase, MapLibre, OpenAI

- Placed 9th of 60+ teams at Hack the Change 2025 with a 311 analytics dashboard built in 24 hours, ingesting 1,000+ Calgary service requests into Supabase behind a community district map and live activity feed
- Designed an LLM analytics assistant that aggregates open requests by community and category server side into a compact structured summary before prompting, keeping token cost flat as the dataset grows instead of streaming raw rows to the model

SKILLS

Languages: Go, Python, TypeScript, JavaScript, Java, C/C++ , SQL

Frameworks: React, Next.js, React Native, StencilJS, TailwindCSS, Webpack, Echo, Encore.dev, Flask

Infrastructure & Testing: AWS, Terraform, PostgreSQL, Supabase, Vercel, Git, Linux, Vitest, Playwright, JUnit